

**GETUIGSCHRIFT HOGER BEROEPSONDERWIJS****DEGREE CERTIFICATE HIGHER EDUCATION**

De voor de bacheloropleiding HBO-ICT ingestelde
examencommissie van Fontys ICT, onderdeel van Fontys
Hogeschool, verklaart dat

The Examination Board, established for the bachelor's
degree programme in Information & Communication
Technology of Fontys Information and Communication
Technology, part of Fontys University of Applied Sciences,
certifies that

Dimitar Stoyanov Barev

geboren op (born) 18 juli (July) 2003

met goed gevolg het afsluitend examen heeft afgelegd
van de bacheloropleiding

successfully completed the exams of the Bachelor's
Degree Programme

HBO-ICT

**Information & Communication
Technology**

Het instellingsbestuur verleent de graad

The Executive Board awards the degree

Bachelor of Science

Tevens heeft betrokkene recht op het voeren van de titel
ingenieur (ing.).

The title obtained with this degree is ingenieur,
abbreviated to ing.

Eindhoven, 2 juli 2026

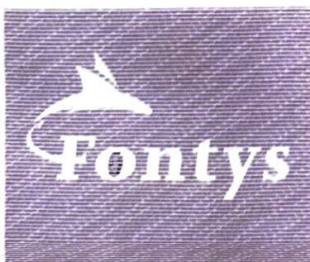
De examencommissie / The Examination Board:

Voorzitter / Chair:
V.Y. Kabzar, MSc

Secretaris / Secretary:
C.M. van Vucht, BSc Hons

De geëxamineerde / The candidate:

4751671-2026-7-006585



Het examen is afgenomen op grond van de Onderwijs- en examenregeling zoals vastgesteld door het instellingsbestuur volgens de eisen in de Wet op het hoger onderwijs en wetenschappelijk onderzoek (WHW, Stb. 1992, 593). De opleiding is geaccrediteerd in juni 2024 en opgenomen in het register Registratie Instellingen en Opleidingen (RIO) onder nummer 30020.

The examination was conducted in virtue of the Education and Examination Regulations as laid down by the Executive Board in accordance with the requirements as decreed in the Higher Education and Research Act (WHW, 1992, Stb. 593). The degree programme was accredited in June, 2024 and is registered in the Institutions and Programmes Register (RIO), number 30020.

De aan de ommezijde vermelde verklaring, betreffende

The certificate concerning

Dimitar Stoyanov Barev

die met goed gevolg het afsluitend examen heeft afgelegd van de bacheloropleiding **HBO-ICT**, heeft betrekking op de volgende onderdelen:

Vrije keuzeruimte: Buitenlandminor; Victoria University, Melbourne - AUSTRALIA

Agile Development of Client-server Applications
Introduction Into Cyber Attack and Defense
Internship ICT
Advanced ICT & Software Engineering
Graduation ICT

who successfully completed the exams of the Bachelor's Degree programme **Information & Communication Technology**, applied to the following units of study:

Free choice space: Studying Abroad; Victoria University - AUSTRALIA

Agile Development of Client-server Applications
Introduction Into Cyber Attack and Defense
Internship ICT
Advanced ICT & Software Engineering
Graduation ICT

4751671-2026-7-006585

Onder andere de volgende elementen kenmerken de echtheid van dit document:

1. Een microtekst in het vierkante paarse vlak van het logo onderaan het document. Als u dit vlak 10 keer of meer vergroot wordt de tekst leesbaar.
2. Een zilverfoliestrip rechts op het document. In de zilverfolie is het monogram van Fontys Hogeschool (de letter 'F' met Fontysvis) uitgespaard.
3. Een uniek nummer links naast het monogram, tevens uitgevoerd als onzichtbare UV-nummering. Bij twijfel over de echtheid van het document kunt u bij Fontys Hogeschool verifiëren of de naam en het nummer corresponderen.
4. In het papier zit een wereldwijd beschermd watermerk.

The purpose of the Diploma Supplement is to provide sufficient independent data to improve the international 'transparency' and fair academic and professional recognition of qualifications (diplomas, degrees, certificates etc.). It is designed to provide a description of the nature, level, context, content and status of the studies that were pursued and successfully completed by the individual named on the original qualification to which this supplement is appended. It is free from any value judgements, equivalence statements or suggestions about recognition. This Diploma Supplement model was developed by the European Commission, Council of Europe and UNESCO.

1. Information identifying the holder of the qualification

- 1.1 Last name(s)** : Barev
1.2 First name(s) : Dimitar Stoyanov
1.3 Date of birth : 18 July 2003
1.4 Student identification number / code : 4751671

2. Information identifying the qualification

- 2.1 Name of qualification and (if applicable) title conferred (in original language):**
 Getuigschrift Hoger Beroepsonderwijs (Degree Certificate of Higher Professional Education);
 Degree: Bachelor of Science;
 Nationally accepted title: Ingenieur (ing.)
- 2.2 Main field(s) of study for the qualification:**
 HBO-ICT (Information & Communication Technology), registered in the "Register Registratie Instellingen en Opleidingen". (RIO) the Institutions and Programmes Register; Registration No. 30020.
- 2.3 Name and status of awarding institution (in original language):**
 Fontys Hogeschool; University of Applied Sciences, state recognized institution, falling within the scope of the Higher Education and Research Act (WHW Stb. 593, 1992 and subsequent amendments and supplements)
- 2.4 Name and status of institution (if different from 2.3) administering studies (in original language):**
 See 2.3
- 2.5 Language of instruction/examination:**
 The Bachelor's Degree programme in Information & Communication Technology is offered in English in Eindhoven and in Dutch in Tilburg. Optional, the first two semesters in Eindhoven are also offered in Dutch. Internships can be either in English or Dutch.

3. Information on the level of the qualification

- 3.1 Level of qualification:**
 Bachelor's Degree; University of Applied Sciences; First cycle National Qualifications Framework for Higher Education; Level 6 EQF/NLQF for LLL.
- 3.2 Official duration of programme in credits and/or years:**
 4 years; study load in ECTS-credits: 240
- 3.3 Access requirement(s):**
 mbo4, havo, vwo; see "Requirements for Access to Higher Education"
 Prospective students with a certificate issued outside the Netherlands, who would like to study in Eindhoven, must additionally demonstrate that their command of the English language is adequate.
 Prospective students with a certificate issued outside the Netherlands, who would like to study in Tilburg, must additionally demonstrate that their command of the Dutch language is adequate.

4. Information on the programme completed and the results obtained

- 4.1 Mode of study:**
 Full-time, including work placements and / or research

4.2 Programme learning outcomes:

Graduates who have successfully completed the Bachelor's Degree programme in Information & Communication Technology have shown proficiency on two sets of learning outcomes.

The first set of learning outcomes concerns the five professional duties on architectural layers of the Dutch national bachelor HBO-i Domain Description: user interaction, organisational processes, infrastructure, software and hardware interfacing. The professional duties are divided into five activities: analysis, advise, design, realise, manage & control. Every graduate has achieved the bachelor's proficiency level (level 3) for these professional duties in one or more architectural layers. For at least four additional activities the post-propedeutic proficiency level (level 2) has been achieved.

The second set concerns the four professional skills of the HBO-i Domain Description: personal leadership, targeted interaction, investigative ability and future oriented-organisation. In short, personal leadership refers to an enterprising attitude, learning capabilities and professional aspirations. Targeted interaction refers to competence in communication and cooperation. Investigative ability refers to research skills in terms of identifying problems, implementing effective approaches and arriving at appropriate solutions. Future oriented-organisation refers to exploring organisational contexts and making business, sustainable and ethical considerations. On the entire set bachelor proficiency (level 3) has been achieved.

4.3 Programme details, individual credits gained and grades/marks obtained (if this information is available in an official transcript this should be used here):

The programme details, individual credits and grades obtained are specified in the appendix Diploma Supplement Transcript of Credits (see below). Students can pass a semester with the scores Satisfactory (S), Good (G) or Outstanding (O). The Bachelor's Degree programme in Information & Communication Technology starts with an orienting and selective first semester on the five architectural layers user interaction, organisational processes, infrastructure, software and hardware interfacing. These are referred to as profiles. The second semester is a further introduction in the profile of choice. Completing this semester entitles the student to a propedeutic certificate. From the second year onwards students have the option to do semesters in their profiles, in specialisations or in personal custom programmes. This means that after the propedeutic year every student has an individual study route. There are two internships. The first is in the fifth semester. The second internship is the final graduation semester. After the first internship students can opt to do a minor inside or outside Fontys University of Applied Sciences.

4.4 Grading system and, if available, grade distribution table:

Grading system in the Netherlands

Dutch grades range from 1 (very poor) to 10 (excellent); a 6 is a pass. It should be noted that 9s and 10s are rarely given. On final lists, grades are normally rounded off (above 0.5 is rounded up and below 0.5 is rounded down, thus a 5.5 equals a 6 equals a pass, whereas a 5.4 equals a fail). The Dutch ten-point grading system is listed in the table below. Related to this national scale the five-point grading scale (shaded rows) and the three-point grading scale) are depicted.

Grade	In words NL	Abbr. NL	Meaning EN	Abbr. EN	Expression NL	Abbr. NL	Expression EN	Abbr. EN
10	Uitmuntend	U	Excellent (*)	E				
9	Zeer Goed	ZG	Very Good	VG				
8	Goed	G	Good	G	Boven Niveau	BN	Above Level	AL
7	Ruim Voldoende	RV	More than Satisfactory	MS				
6	Voldoende	V	Satisfactory	S	Op Niveau	N	At Level	L
5	Bijna Voldoende	BV	Almost Satisfactory	AS				
4	Onvoldoende	O	Unsatisfactory	U	Onder Niveau	ON	Below Level	BL
3	Zeer Onvoldoende	ZO	Very Unsatisfactory	VU				
2	Slecht	S	Poor	P				
1	Zeer Slecht	ZS	Very Poor	VP				
*) displays as Outstanding (O) using the ten-point word mark scale					Behaald	BEH	Pass	PA
					Niet Behaald	NB	Fail	F
					Vrijstelling	VR	Exemption	EX

Grade Distribution Table

Study programme: Bachelor Information & Communication Technology , register number 30020					
Reference group: 3 previous academic years 2021-2022 - 2023-2024, number of passing grades: 16789					
Fontys School: Fontys Information and Communication Technology					
Grade:	5.5-6.4	6.5-7.4	7.5-8.4	8.5-9.4	9.5-10
%	43.0%	3.0%	37.0%	2.0%	15.0%
Cum. %	100.0%	57.0%	54.0%	17.0%	15.0%

See: <https://www.fontys.nl/GradeDistribution> for information on the Grade Distribution at Fontys Hogeschool.

4.5 Overall classification of the qualification (in original language):

met goed gevolg (Successfully)

5. Information on the function of the qualification

5.1 Access to further study:

The Degree Certificate Higher Education gives access to several types of further study: Related master degree programmes; Shortened university study programmes leading to the university Degree Certificate; Post graduate courses of varying length.

5.2 Professional status:

HBO graduates are fully qualified to practise their profession on the bachelor level without further study, licensing or registration with a professional association.

6. Additional information

6.1 Additional information:

The degree programme was accredited by the Accreditation Organization of the Netherlands and Flanders (Nederlands-Vlaamse Accreditatie Organisatie, NVAO) on 11 June 2024.

6.2 Further information sources:

For general information about Fontys: <http://www.fontys.nl>

For information in the English language: <http://fontys.edu/about-fontys>

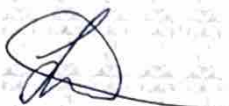
Further information on Higher Professional Education in the Netherlands can be obtained at: Nuffic, Kortenaerkade 11, P.O. Box 29777, 2502 LT Den Haag, The Netherlands: <http://www.epnuffic.nl>

There is also an important site on the Recognition of Diplomas: <http://ec.europa.eu/education>

7. Certification of the supplement

7.1 Date : 2 July 2026

7.2 Signature :



7.3 Capacity : V.Y. Kabzar, MSc ; Chair of the Examination Board

7.4 Official stamp or seal :



8. Information on the Dutch higher education system: See enclosed information

Diploma Supplement Transcript of Credits (see No. 4.2/4.3.)

DEGREE PROGRAMME: Bachelor Information & Communication Technology
(registered name: Bachelor HBO-ICT)

Last name(s) : Barev
First name(s) : Dimitar Stoyanov
Date of birth (day/month/year) : 18/07/2003
Date of qualification : 2 July 2026
Student identification number : 4751671
Certificate identification number : 4751671-2026-7-006585

Free choice space : Subject name : Buitenlandminor; Victoria University, Melbourne - AUSTRALIA
Subject name in English : Studying Abroad; Victoria University - AUSTRALIA

Course components

Code	Subject name	Subject name in English	Date	Grades	ECTS
	<u>Propedeusefase</u>	<u>Propaedeutic Phase</u>			
	<u>Semester 1</u>	<u>Semester 1</u>			
29V1C	Oriëntatie op ICT	Orientation on ICT	30/01/2023	G	30
	<u>Semester 2 ICT & Software Engineering</u>	<u>Semester 2 ICT & Software Engineering</u>			
29VS2A	Basic Software Engineering	Basic Software Engineering	05/07/2023	S	30
	<u>Hoofdfase</u>	<u>Main Phase</u>			
	<u>Semester 3 ICT & Software Engineering</u>	<u>Semester 3 ICT & Software Engineering</u>			
29VS3B	Agile Development of Client-server Applications	Agile Development of Client-server Applications	06/02/2024	S	30
	<u>Semester 4 ICT & Cyber Security</u>	<u>Semester 4 ICT & Cyber Security</u>			
29VCS4	Introduction Into Cyber Attack and Defense	Introduction Into Cyber Attack and Defense	05/07/2024	S	30
	<u>Semester 5</u>	<u>Semester 5</u>			
29V5	Internship ICT	Internship ICT	28/01/2025	G	30
	<u>Semester 6</u>	<u>Semester 6</u>			
29VSA-H	Advanced ICT & Software Engineering	Advanced ICT & Software Engineering	07/07/2025	S	30
29VSA	Enterprise Software Engineering	Enterprise Software Engineering	07/07/2025	S	30
	<u>Semester 7</u>	<u>Semester 7</u>			
2557FMBUIT	Buitenlandminor; Victoria University, Melbourne - AUSTRALIA	Studying Abroad; Victoria University - AUSTRALIA	08/01/2026	PA	30
	<u>Semester 8</u>	<u>Semester 8</u>			
29-TITEL	Benchmarking OCR Architectures for Intelligent Insurance Invoice Processing	Benchmarking OCR Architectures for Intelligent Insurance Invoice Processing			
29V8-H	Graduation ICT	Graduation ICT	02/07/2026	G	30
29V8A	Assignment	Assignment	02/07/2026	G	30
	Gewogen gemiddelde / Totaal aantal behaalde ECTS-studiepunten	Weighted average / Total number of ECTS-credits achieved		6.9	240

Diploma Supplement Appendix Competences

Description of Architectural Layers

User Interaction (Ui)	The 'user interaction' architectural layer relates to the communication between the user or end user and the ICT system. This emphatically does not refer to the interaction with users, as it takes place during the creation of an ICT system; after all, this takes place in each of the architectural layers.
Organisational Processes (Op)	The 'organisational processes' architectural layer relates to the facilitation of organisational processes by means of ICT systems. This involves the functionality of the system as a whole (automated and non-automated components), viewed from the context of organisational goals to be realised.
Infrastructure (In)	The 'infrastructure' architectural layer concerns the entirety of ICT-systems used to facilitate organisational processes. This involves the making available, keeping available and configuring the traditional hardware infrastructure, and certainly also the software infrastructure.
Software (So)	The 'software' architectural layer concerns the development of various kinds of software. This concerns software that can be incorporated into an ICT infrastructure after delivery.
Hardware Interfacing (Hi)	The 'hardware interfacing' architectural layer concerns software that interacts with available hardware. This involves situations where the software must explicitly take into account the capabilities and limitations of the available hardware. In the description within this architectural layer, 'system' has been chosen as the generic, overarching term. Depending on the context, this can be further specified as 'embedded system', 'industrial automation', 'virtual system', etc.

Description of Activities

Analysis (An)	The 'analysis' activity involves the analysis of processes, products and information flows in their interrelationship and context.
Advise (Ad)	The 'advise' activity involves advising on the organisation of processes and/or information for an ICT system to be newly developed purchased or modified.
Design (De)	The 'design' activity involves designing an ICT system (or part thereof) based on requirements.
Realise (Re)	The 'realise' activity involves creating and testing an ICT system (or part thereof) based on a design.
Manage & Control (MC)	The 'manage & control' activity involves managing, monitoring and optimising the development, deployment and use of ICT systems.

Proficiency Levels

Level	Autonomy	Complexity Context	Complexity Content
1. Task-Oriented	Works under general guidelines in an environment where unpredictable changes take place.	Structured - predictable context, problem defined, approach and solution known to client.	A few basic concepts that build on prior education.
2. Problem-oriented	Autonomously solves interactive issues that arise from project activities.	Structured - unpredictable context, problem given, choice of approach and space for solutions limited.	Combination of several basic concepts and a few in-depth concepts that build on basic concepts.
3. Situation-oriented	Works autonomously to solve interactive problems. Has a positive effect on team performance.	Structured - unpredictable context, vague problems, approach and solution space open.	Combination of several concepts for delving deeper into and innovation in the local situation.

Applying Knowledge and Understanding

	An	De	Re	Ad	McC
Ui			2		
Op	2	2			
In		2	2		
So	3	3	3	3	3
Hi					

Performance Indicators

Criteria for Professional Skills

Future-oriented Organisation	Within future-oriented organisation, the environment in which ICT work takes place is key. The focus here is on the working method in relation to the assignment and the context factors that influence the work and/or the products to be developed or further developed.
Investigative Ability	Within investigative ability, questions related to the assignment are key. The emphasis in this theme is on critical thinking and working methodically with substantiated choices.
Personal Leadership	Personal leadership focuses on the development of the individual in relation to the context in which he/she operates. This includes attention to proactive action based on information obtained in order to shape further personal and professional development.
Targeted Interaction	Communication and collaboration play a key role in targeted interaction. Both the form and content of communication and collaboration aspects are important in this.